

Covered: Properties of Adjoint Operator, Similarity of Self-Adjoint/Hermitian Matrix, Orthogonal Basis of Eigenvalues of Hermitian Operator (for finite dimensional IPVS), Definition of Unitary Operator, and its Equivalent Definitions

Remark 0.1. A natural question arises: if $A, B \in M_n(\mathbb{C})$ such that A is Self-Adjoint/Hermitian and similar to B , is B necessarily Self-Adjoint/Hermitian as well? If the answer is affirmative, we could extend our analysis to the entire equivalence class of similar matrices, treating any member of this class as self-adjoint, similar to the case when we exploit properties that are invariant under similarity, diagonalizability, characteristic polynomial, minimal polynomial, triangularizable etc.

If $A, B \in M_n(\mathbb{C})$ such that A is Self-Adjoint/Hermitian and similar to B , there exists an invertible matrix $P \in M_n(\mathbb{C})$ such that $B = P^{-1}AP$. $B^* = P^*A(P^{-1})^*$, $B = B^*$ need not be true in general if $P^{-1} \neq P^*$. Take

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 4 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\text{then } B = \begin{bmatrix} 0 & -3 \\ 1 & 5 \end{bmatrix} \quad \text{and} \quad B^* = \begin{bmatrix} 0 & 1 \\ -3 & 5 \end{bmatrix} \quad \text{and} \quad B \neq B^*$$

Remark 0.2. The above example and argument motivates us to study matrices and operator with the property that, $P^{-1} = P^*$, which will be done shortly.

Lemma 0.3. Let $A \in M_n(\mathbb{F})$ we have $(A^*)^* = A$.

Proof. Let $x, y \in \mathbb{F}^n$,

$$\langle Ax, y \rangle = \langle x, A^*y \rangle = \overline{\langle A^*y, x \rangle} = \overline{\langle y, (A^*)^*x \rangle} = \langle (A^*)^*x, y \rangle$$

$$\implies (A^*)^* = A.$$

Alternate proof,

$$\langle A^*x, y \rangle = \overline{\langle y, A^*x \rangle} = \overline{\langle A^*y, x \rangle} = \langle x, Ay \rangle, \quad \forall x, y \in \mathbb{F}_n,$$

By the uniqueness of adjoint, $(A^*)^* = A$. □

Theorem 0.4. Let $A \in L(V)$ (can be matrix or operator), where V is finite dimensional inner product vector space and W be subspace of V . W is an A -invariant subspace of V if and only if $W^\perp$ is an A^* -invariant subspace of V .

Proof. $[\implies]$ Assume W is an A -invariant subspace of V . Let $y \in W^\perp$, then

$$\forall z \in W \quad \langle A^*y, z \rangle = \langle y, (A^*)^*z \rangle = \langle y, Az \rangle = 0$$

Thus $A^*y \perp z, \forall z \in W$, i.e., $A^*y \in W^\perp$. Hence $W^\perp$ is an A^* -invariant subspace of V .

[$\Rightarrow$] Conversely assume that $W^\perp$ is an A^* -invariant subspace of V .

Then by above result we know that $(W^\perp)^\perp$ is an $(A^*)^*$ -invariant subspace, i.e. W is an A -invariant subspace of V . $\square$

Theorem 0.5. *Let $A \in L(V, W)$, then (works for infinite dimensional space as well)*

1. $\text{Range } A^* = (\text{Ker}(A))^\perp$

2. $\text{Range } A = (\text{Ker}(A^*))^\perp$

3. $\text{Ker}(A^*) = (\text{Range } A)^\perp$

4. $\text{Ker}(A) = (\text{Range } A^*)^\perp$

Proof. (1) Firstly, let $A^*x \in R(A^*)$ (denoting Range A as $R(A)$). To show $\forall y \in \text{Ker } A, A^*x \perp y$.

Let $y \in \text{Ker } A$, then $Ay = 0 \iff \langle x, Ay \rangle = 0 \iff \langle A^*x, y \rangle = 0$.

Thus $\forall y \in \text{Ker } A, A^*x \perp y \implies A^*x \in \text{Ker } A \implies R(A^*) \subseteq (\text{Ker}(A))^\perp$.

Secondly, to show $(\text{Ker}(A))^\perp \subseteq R(A^*)$, equivalently, $(\text{Ker}(A)) \subseteq (R(A^*))^\perp$.

Let $y \in \text{Ker}(A), Ay = 0$. For $x \in W$

Rest can be proved using above relation along with $(W^\perp)^\perp = W \& (A^*)^* = A$. $\square$

Question: Is it true that Range A is closed subset? Is it closed for Infinite Dimensional IPVS?

Proposition 0.6. *For a Finite Dimensional Inner Product Vector Space $(V, \langle \cdot, \cdot \rangle)$ over Field $\mathbb{F}$, any subspace W is closed subset of $(V, \|\cdot\|)$.*

Proof. Let $\{v_1, v_2, \dots, v_k\}$ be basis of W and $\{v_1, v_2, \dots, v_n\}$ be extended basis of V . Let x_n be sequence in W converging to $x \in V$, then $x_n = \sum_1^n \alpha_{in} v_i = \sum_1^k \alpha_{in} v_i$ and $x = \sum_1^n \langle x, v_i \rangle v_i$. Since $x_n \rightarrow x$, we have $\alpha_{in} \rightarrow \langle x, v_i \rangle$, for $1 \leq i \leq k$. But for $j > k, \alpha_{jn} = 0 \implies \langle x, v_j \rangle = 0 \implies x \in W$

Alternate Proof: W is subspace of V , and thus W is finite dimensional space. Recall: Every Finite Dimensional Normed Space is Complete. Thus W is complete, also V is complete, hence W is closed.

Remark 0.7. *Subspace of Arbitrary space need not be closed. But Orthogonal Complement is always Closed.*

1 Hermitian/Self-Adjoint

Proposition 1.1. *Let $(V, \langle \cdot, \cdot \rangle)$ be IPVS over Field $\mathbb{C}$. Let $A \in L(V)$ be Hermitian/Self-Adjoint, then All Eigenvalues of A are Real.*

Proof. Given $A \in L(V)$ be Hermitian. All the eigen values of A exists. Let $\lambda \in \mathbb{C}, x \in V, x \neq 0$, such that $Ax = \lambda x$. Consider

$$\lambda \langle x, x \rangle = \langle \lambda x, x \rangle = \langle Ax, x \rangle = \langle x, A^* x \rangle = \langle x, Ax \rangle = \overline{\lambda} \langle x, x \rangle$$

Since $x \neq 0$, then $\lambda = \overline{\lambda}$, implies λ is real.

Remark 1.2. Question arises if $A \in L(V)$, such that $\langle Ax, x \rangle$ is real then $A = A^*$?

Take $A = \begin{bmatrix} 15 & 5 \\ 1 & 29 \end{bmatrix}$ then $\langle Ax, x \rangle$ is real for any vector x if V is considered over $\mathbb{R}$, but A is not Hermitian.

Theorem 1.3. Let V be IPVS over field $\mathbb{C}$, and A be linear operator on V . A is Hermitian if and only if $\langle Ax, x \rangle$ is real for all vectors x in V .

Proof. Given that $A = A^*$, let $x \in V$.

$$\text{Say } \beta = \langle Ax, x \rangle = \langle x, A^* x \rangle = \langle x, Ax \rangle = \overline{\langle Ax, x \rangle} = \overline{\beta}$$

Thus $\langle Ax, x \rangle$ is real for all $x \in V$.

Recall Polarization Identity,

$$\langle x, y \rangle = \frac{1}{4} \left(\sum_{k=0}^3 i^k \langle x + i^k y, x + i^k y \rangle \right) \quad \forall x, y \in V$$

$$\text{then } \langle Ax, y \rangle = \frac{1}{4} \left(\sum_{k=0}^3 i^k \langle Ax + i^k y, Ax + i^k y \rangle \right) \quad \forall x, y \in V$$

From calculations we can conclude that $\langle Ax, y \rangle = \overline{\langle Ax, y \rangle} = \langle y, Ax \rangle = \langle A^* x, y \rangle$.

Hence $\langle (A - A^*)x, y \rangle = 0, \forall x, y \in V \implies A = A^*$.

Alternate Proof for reverse direction, $\langle Ax, x \rangle = \overline{\langle Ax, x \rangle} = \langle x, Ax \rangle = \langle A^* x, x \rangle$.

Then $\langle (A - A^*)x, x \rangle = 0, \forall x \in V$, hence $A = A^*$.

Proposition 1.4. Let A be Hermitian Linear Operator on V and x, y be eigenvectors of A corresponding to λ, μ , two distinct eigenvalues. Then $\langle x, y \rangle = 0$.

Proof. Given that $Ax = \lambda x, Ay = \mu y, x \neq 0, y \neq 0$.

$$\lambda \langle x, y \rangle = \langle Ax, y \rangle = \langle x, Ay \rangle = \mu \langle x, y \rangle$$

$$\lambda \neq \mu \implies \langle x, y \rangle = 0.$$

Theorem 1.5 (Spectral Theorem for Hermitian Operators). Let V be a finite-dimensional inner product space and let $A \in \mathcal{L}(V)$ be a Hermitian (self-adjoint) operator. Then there exists an orthonormal basis $\mathcal{B}$ of V such that the matrix representation $[A]_{\mathcal{B}}$ is diagonal. Equivalently, there exists an orthonormal basis of V consisting of eigenvectors of A .

Proof. We proceed by Principal of Mathematical Induction on $n = \dim V$. If $n=1$, choose a unit vector $v_1 \in V$. Then $\mathcal{B} = \{v_1\}$ is an orthonormal basis for V . Since $Av_1 = \lambda v_1$ for some scalar λ , the matrix representation $[A]_{\mathcal{B}} = [\lambda]$ is diagonal.

Assume that, statement holds for any Hermitian operator on an inner product space of dimension $n - 1$.

To prove statement when $\dim V = n$, Let λ be an eigenvalue of A , and let $v_1 \in V$ be an eigenvector with $\|v_1\| = 1$, thus $Av_1 = \lambda v_1$. Define $W_1 = \text{span}\{v_1\}$. Since v_1 is an eigenvector, W_1 is A -invariant. Since V is a finite-dimensional inner product space,

$$V = W_1 \oplus W_1^\perp$$

Because A is Hermitian ($A^* = A$) and W_1 is A -invariant, $W_1^\perp$ is invariant under $A^* = A$. For any $x \in W_1^\perp$ and $cv_1 \in W_1$,

$$\langle Ax, cv_1 \rangle = \langle x, A^*(cv_1) \rangle = \langle x, A(cv_1) \rangle = \langle x, c\lambda v_1 \rangle = 0$$

which implies $Ax \in W_1^\perp$.

Consider the operator $A_2 = A|_{W_1^\perp} : W_1^\perp \rightarrow W_1^\perp$. A_2 is self-adjoint on $W_1^\perp$ since $\forall x, y \in W_1^\perp$,

$$\langle A_2x, y \rangle = \langle Ax, y \rangle = \langle x, A^*y \rangle = \langle x, Ay \rangle = \langle x, A_2y \rangle \implies A_2^* = A_2$$

Since $\dim W_1^\perp = n - 1$, by the inductive hypothesis, there exists an orthonormal basis $\mathcal{B}' = \{v_2, \dots, v_n\}$ of $W_1^\perp$ consisting of eigenvectors of A_2 such that:

$$[A_2]_{\mathcal{B}'} = \text{diag}(\mu_1, \mu_2, \dots, \mu_{n-1})$$

Now take $\mathcal{B} = \{v_1\} \cup \mathcal{B}' = \{v_1, v_2, \dots, v_n\}$. Since $v_1 \perp W_1^\perp$ and $\mathcal{B}'$ is orthonormal, $\mathcal{B}$ is an orthonormal basis for V consisting of eigenvectors of A . The matrix representation $[A]_{\mathcal{B}}$ is:

$$[A]_{\mathcal{B}} = \begin{bmatrix} \lambda & 0 \\ 0 & [A_2]_{\mathcal{B}'} \end{bmatrix} = \begin{bmatrix} \lambda & 0 & \cdots & 0 \\ 0 & \mu_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mu_{n-1} \end{bmatrix}$$

which is diagonal. By induction, the result holds for all $n \geq 1$. $\square$

Definition 1.6 (Unitary Operator/Matrix). Let V be Inner Product Vector Space over field $\mathbb{F}$. The linear operator U on V (or matrix A on V) is called Unitary Operator (or unitary matrix) if $UU^* = I = U^*U$ (or $AA^* = I = A^*A$).

Theorem 1.7. Let $U \in L(V)$ (operator or matrix) over field $\mathbb{F}$. Then the followings are equivalent:

1. U is Unitary Operator, i.e., $UU^* = I = U^*U$.

2. $\forall x, y \in V, \langle Ux, Uy \rangle = \langle x, y \rangle$.
3. $\forall x \in V, \|Ux\| = \|x\|$.
4. U sends one Orthonormal Basis of V to another Orthonormal Basis of V .
5. U sends every Orthonormal Basis of V to every Orthonormal Basis of V .

Proof. 1 $\rightarrow$ 2. $\forall x, y \in V, \langle Ux, Uy \rangle = \langle x, U^*Uy \rangle = \langle x, y \rangle$.

2 $\rightarrow$ 3. $\forall x \in V, \|Ux\| = \|x\|$.

3 $\rightarrow$ 2. From Polarization Identity, we have, $\forall x, y \in V$

$$\begin{aligned} \langle Ux, Uy \rangle &= \frac{1}{4} \sum_{k=0}^3 i^k \|Ux + i^k Uy\|^2 = \frac{1}{4} \sum_{k=0}^3 i^k \|U(x + i^k y)\|^2 \\ &= \frac{1}{4} \sum_{k=0}^3 i^k \|x + i^k y\|^2 \\ &= \langle x, y \rangle \end{aligned}$$

2 $\rightarrow$ 5. Let $\{v_1, v_2, \dots, v_n\}$ be orthonormal basis of V . Recall U is invertible, thus $\{Uv_1, Uv_2, \dots, Uv_n\}$ is basis of V .

$$\langle Uv_i, Uv_j \rangle = \langle v_i, U^*Uv_j \rangle = \langle v_i, v_j \rangle = \delta_{ij}.$$

Thus $\{Uv_1, Uv_2, \dots, Uv_n\}$ is Orthonormal Basis of V .

5 $\rightarrow$ 4 is obvious.

4 $\rightarrow$ 1. Let $\{u_1, u_2, \dots, u_n\}$ be one orthonormal basis of V which is mapped to orthonormal basis $\{Uu_1, Uu_2, \dots, Uu_n\}$ of V .

$$\langle U^*Uu_i, u_j \rangle = \langle Uu_i, Uu_j \rangle = \delta_{ij} = \langle u_i, u_j \rangle$$

$$\forall 1 \leq i, j \leq n, \langle (U^*U - I)u_i, u_j \rangle = 0 \implies U^*U = I$$

Hence, U is unitary operator (or matrix). $\square$

Theorem 1.8. Let $(V, \langle \cdot, \cdot \rangle_V)$ and $(W, \langle \cdot, \cdot \rangle_W)$ be finite-dimensional inner product spaces over the same field, having the same dimension. If $T \in L(V, W)$, the following are equivalent:

1. U is Unitary Operator, i.e., $TT^* = I_W, T^*T = I_V$.
2. $\forall x, y \in V, \langle Tx, Ty \rangle_W = \langle x, y \rangle_V$.
3. $\forall x \in V, \|Tx\|_W = \|x\|_V$.
4. T sends one Orthonormal Basis of V to another Orthonormal Basis of W .
5. T sends every Orthonormal Basis of V to every Orthonormal Basis of W .

Definition 1.9. A and B be linear operator on V (or matrices) are congruent if there exists $X \in L(V)$ such that $B = X^*AX$

Remark 1.10. *Note that in above definition X is not considered to be invertible, but in our course, we will mostly work with invertible matrices. Similarity by unitary operator and Congruence by unitary operator are two different concepts. Inner Product Structure works well under change of orthonormal basis, rather than Linearly Independent Basis.*